

# Expand on paper claims



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PDF

Hello, I would like you to expand on the arguments given for the claims in a paper I am reading (A Spectral Condition ...). When doing so, could you please make sure your explanation is:

- A. Precise
- B. Easy to follow: proofs should be in English prose as much as possible (while still being precise)
- C. Self-Contained: Please try to only reference facts a CS182 student would know. If you need to include new facts/arguments, please state them in full before using them
- D. Well structured

Also, at the end, please try to generate some questions to check conceptual understanding

Here's a structured, self-contained expansion of the arguments around **Claim 1, Claim 2, and their use in bounding hidden feature and update norms** from the excerpt you gave. □

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## 0. Quick recap of the setting

ChatGPT Pro in general performed very well. The explanations (for me) were (largely) clear, correct, and self-contained.

We are looking at one hidden layer of a deep network.

- Input to this layer: a feature vector  $h_{\ell-1}(x) \in \mathbb{R}^{n_{\ell-1}}$ .
- Weight matrix of this layer:  $W_{\ell} \in \mathbb{R}^{n_{\ell} \times n_{\ell-1}}$ .
- Output of this layer (before nonlinearity):

$$h_{\ell}(x) = W_{\ell} h_{\ell-1}(x).$$

They are interested in how large  $\|h_{\ell}(x)\|_2$  and  $\|\Delta h_{\ell}(x)\|_2$  are, and how these scale with the layer width  $n_{\ell}$ .

A key linear algebra fact being used is **submultiplicativity of the spectral norm**:

For any matrix  $A$  and vector  $v$ ,

One of the few things I didn't like was the choice of notation for the spectral norm of a matrix. Here  $\|A\|_{\rightarrow}$  actually means the spectral norm of  $A$

$$\|Av\|_2 \leq \|A\|_{\rightarrow} \|v\|_2,$$

where  $\|A\|_{\rightarrow}$  is the spectral norm of  $A$  (the largest singular value).

In general, this inequality can be very loose. The section is arguing that for deep nets under typical conditions, it is *not* loose:  $\|Av\|$  is actually on the order of  $\|A\| \cdot \|v\|$  (i.e., they're "aligned").

## 1. Claim 1 – Alignment of initial weight matrices

### Claim 1 (restated in simpler notation).

Let  $W \in \mathbb{R}^{m \times d}$  be the weight matrix of a hidden layer with fan-in  $d$  and fan-out  $m$ , where  $m \geq d$ . Assume  $W$  is initialized using a standard scheme:

- either “semi-orthogonal” initialization, or
- i.i.d. Gaussian entries with mean 0 and some standard deviation  $\sigma$ .

Then for a fixed input vector  $h \in \mathbb{R}^d$ , with high probability:

$$\|Wh\|_2 = \Theta(\|W\|_{\rightarrow} \cdot \|h\|_2).$$

In words: at initialization, applying  $W$  to  $h$  scales its norm by a factor that is on the same order as the spectral norm of  $W$ .

We now unpack the two cases they mention.

## 1.1. Case 1: semi-orthogonal initialization

### Definition (semi-orthogonal matrix, as used here).

A matrix  $W \in \mathbb{R}^{m \times d}$  with  $m \geq d$  is *semi-orthogonal* (w.r.t. its columns) if its columns are orthonormal; i.e.

$$W^{\top} W = I_d.$$

Equivalently, each column has norm 1 and different columns are orthogonal.

### Two basic consequences:

#### 1. Norm preservation on the column space.

For any  $h \in \mathbb{R}^d$ :

$$\|Wh\|_2^2 = h^{\top} W^{\top} W h = h^{\top} I_d h = \|h\|_2^2 \Rightarrow \|Wh\|_2 = \|h\|_2.$$

#### 2. Spectral norm.

The explanation is still clear and easy to understand. The only complaint is that 1.1 is a little overexplained, ChatGPT could have stoped here

The singular values of  $W$  are the square roots of eigenvalues of  $W^{\top} W$ . Since  $W^{\top} W = I_d$ , all singular values of  $W$  equal 1. So

$$\|W\|_{\rightarrow} = 1.$$

Thus, for any  $h$ ,

$$\|Wh\|_2 = \|h\|_2 = \|W\|_{\rightarrow} \cdot \|h\|_2.$$

This gives **exact equality** for Claim 1 in the semi-orthogonal case.

## 1.2. Case 2: Gaussian initialization

Now assume

$$W_{ij} \sim \mathcal{N}(0, \sigma^2) \quad \text{i.i.d.,}$$

for some  $\sigma = \sigma_\ell$ . We show that:

1.  $\|Wh\|_2$  concentrates around  $\sigma\sqrt{m}\|h\|_2$ .
2.  $\|W\|_{\rightarrow}$  concentrates around  $\sigma(\sqrt{m} + \sqrt{d})$ .
3. Therefore, the ratio  $\frac{\|Wh\|_2}{\|W\|_{\rightarrow}\|h\|_2}$  is  $\Theta(1)$  when  $m \geq d$ .

### 1.2.1. Size of $Wh$

Again, well structured and clear.

Fix a deterministic vector  $h \in \mathbb{R}^d$ . Look at one coordinate of  $Wh$ , say  $(Wh)_i$ :

$$(Wh)_i = \sum_{j=1}^d W_{ij}h_j.$$

Each term  $W_{ij}h_j$  is Gaussian  $\mathcal{N}(0, \sigma^2 h_j^2)$ . Since the entries of  $W$  are independent and the sum of independent Gaussians is Gaussian: My only wish is that one phrase was added here, reminding us that  $W_{ij} \sim \mathcal{N}(0, 1)$ . This is I think personal taste though

$$(Wh)_i \sim \mathcal{N}\left(0, \sigma^2 \sum_{j=1}^d h_j^2\right) = \mathcal{N}(0, \sigma^2 \|h\|_2^2).$$

So each coordinate has variance  $\sigma^2 \|h\|_2^2$ , and they're independent across  $i$  (because different rows of  $W$  are independent).

Then

$$\|Wh\|_2^2 = \sum_{i=1}^m (Wh)_i^2.$$

Each  $(Wh)_i^2$  has expectation  $\sigma^2 \|h\|_2^2$ , so

$$\mathbb{E} \|Wh\|_2^2 = m\sigma^2 \|h\|_2^2.$$

By the **law of large numbers** and standard concentration bounds, for large  $m$ ,

$$\|Wh\|_2^2 \approx m\sigma^2 \|h\|_2^2 \quad \Rightarrow \quad \|Wh\|_2 \approx \sigma\sqrt{m}\|h\|_2$$

with high probability.

So we can write

$$\|Wh\|_2 = \Theta(\sigma\sqrt{m}\|h\|_2)$$

with high probability.

### 1.2.2. Size of $\|W\|_{\rightarrow}$

A standard result from random matrix theory (you don't prove this in CS182, you'd just accept it as a theorem) says:

For an  $m \times d$  Gaussian matrix with entries  $\mathcal{N}(0, \sigma^2)$ ,

$$\|W\|_{\rightarrow} \approx \sigma(\sqrt{m} + \sqrt{d})$$

with high probability, as  $m, d$  grow.

So, up to constants,

$$\|W\|_{\rightarrow} = \Theta(\sigma(\sqrt{m} + \sqrt{d})).$$

### 1.2.3. Putting them together

We have:

$$\|Wh\|_2 \approx \sigma\sqrt{m} \|h\|_2, \quad \|W\|_{\rightarrow} \approx \sigma(\sqrt{m} + \sqrt{d}).$$

Thus

$$\frac{\|Wh\|_2}{\|W\|_{\rightarrow} \|h\|_2} \approx \frac{\sigma\sqrt{m}}{\sigma(\sqrt{m} + \sqrt{d})} = \frac{\sqrt{m}}{\sqrt{m} + \sqrt{d}}.$$

If  $m \geq d$  (fan-out  $\geq$  fan-in), then  $\sqrt{m} \geq \sqrt{d}$ , so

$$\frac{\sqrt{m}}{\sqrt{m} + \sqrt{d}} \in \left[\frac{1}{2}, 1\right].$$

So this ratio is bounded between constant factors; in particular, it does not go to 0 as the dimensions grow. Therefore:

$$\|Wh\|_2 = \Theta(\|W\|_{\rightarrow} \cdot \|h\|_2)$$

with high probability, which is exactly the statement of Claim 1 for Gaussian initialization.

## 2. Claim 2 – Alignment of updates

### Claim 2 (restated).

For a single SGD step with batch size 1 at layer  $\ell$ , the weight update  $\Delta W_{\ell}$  is rank-one and has the previous layer's activations  $h_{\ell-1}(x)$  as its right singular vector. Consequently,

$$\|\Delta W_{\ell} h_{\ell-1}(x)\|_2 = \|\Delta W_{\ell}\|_{\rightarrow} \|h_{\ell-1}(x)\|_2.$$

So for updates, the submultiplicative inequality is actually an **equality**.

### 2.1. Form of the gradient update

Consider a standard fully-connected layer: I found this explanation mostly clear, I think it would have been clearer if it explicitly applied general result about outer-products that it discusses.

$$h_{\ell} = W_{\ell} h_{\ell-1}.$$

Let the loss (on this single example) be  $L(x)$ . Backprop tells us that:

$$\frac{\partial L}{\partial W_{\ell}} = \delta_{\ell}(x) h_{\ell-1}(x)^{\top}$$

for some vector  $\delta_\ell(x) \in \mathbb{R}^{n_\ell}$  of “error signals” at that layer.

For a gradient step with learning rate  $\eta$ ,

$$\Delta W_\ell = -\eta \frac{\partial L}{\partial W_\ell} = -\eta \delta_\ell(x) h_{\ell-1}(x)^\top.$$

The paper writes this generically as

I also like that it brought in notation from the paper. I think this makes it easier to read this paper, and future papers, on your own compared to a completely independent explanation

$$\Delta W_\ell = \alpha_\ell u_\ell(x) h_{\ell-1}(x)^\top,$$

where  $\alpha_\ell$  is a scalar (like  $-\eta$ ) and  $u_\ell(x)$  corresponds to  $\delta_\ell(x)$ . The important point:

**$\Delta W_\ell$  is an outer product of two vectors.**

Therefore, it is a rank-one matrix (assuming neither vector is zero).

So:

- left factor  $u := \alpha_\ell u_\ell(x)$ ,
- right factor  $v := h_{\ell-1}(x)$ ,  
and  $\Delta W_\ell = uv^\top$ .

## 2.2. Spectral norm and action on its right singular vector

For **any** rank-one matrix  $A = uv^\top$ :

- Its only nonzero singular value is

$$\sigma_1(A) = \|u\|_2 \|v\|_2.$$

(This is standard: you can check that  $A^\top A = (vu^\top)(uv^\top) = (u^\top u)(vv^\top)$ , so its only nonzero eigenvalue is  $\|u\|_2^2 \|v\|_2^2$ . Taking square roots gives  $\sigma_1$ .)

- So the spectral norm is

$$\|A\|_{\rightarrow} = \|u\|_2 \|v\|_2.$$

- The **right singular vector** corresponding to this singular value is  $v/\|v\|_2$ . In particular

$$Av = u(v^\top v) = u \|v\|_2^2.$$

Taking norms:

$$\|Av\|_2 = \|u\|_2 \|v\|_2^2.$$

And since  $\|A\|_{\rightarrow} = \|u\|_2 \|v\|_2$ , we get:

$$\|Av\|_2 = \|A\|_{\rightarrow} \|v\|_2.$$

Now plug back  $A = \Delta W_\ell$  and  $v = h_{\ell-1}(x)$ . We obtain:

$$\|\Delta W_\ell h_{\ell-1}(x)\|_2 = \|\Delta W_\ell\|_{\rightarrow} \|h_{\ell-1}(x)\|_2.$$

That's exactly Claim 2. No dimensionality assumptions needed here: it holds purely because of the rank-one outer-product structure.

### 3. Using the claims to bound feature and update norms

The text then uses these claims to argue about the **scaling** of feature norms  $\|h_\ell(x)\|_2$  and update norms  $\|\Delta h_\ell(x)\|_2$  as layer widths grow. I'll sketch this at a conceptual level.

#### 3.1. Initial feature norms

This section is not (imo) super necessary given the conceptual overview in the beginning, but might be useful to repeat in more mathematical terms

At initialization,

$$h_\ell(x) = W_\ell h_{\ell-1}(x).$$

Claim 1 tells us:

$$\|h_\ell(x)\|_2 = \Theta(\|W_\ell\|_{\rightarrow} \cdot \|h_{\ell-1}(x)\|_2).$$

They also refer to "Condition 1", which (loosely) encodes properties of the initialization (e.g., how the spectral norm scales with width). Combining induction on layers with this condition, they argue that

$$\|h_\ell(x)\|_2 = \Theta(\sqrt{n_\ell}),$$

i.e. feature norms scale like the square root of the layer width (this is the "correct" scaling they want).

#### 3.2. Update of activations $\Delta h_\ell(x)$

From the chain rule,

$$\Delta h_\ell(x) = \Delta W_\ell h_{\ell-1}(x) + W_\ell \Delta h_{\ell-1}(x) + \Delta W_\ell \Delta h_{\ell-1}(x).$$

- The first term  $\Delta W_\ell h_{\ell-1}(x)$  corresponds to the update in this layer due to the current weights changing.
- The second term propagates earlier changes up.
- The third term is a second-order term (product of two small changes); they keep it but then argue about its relative size.

By Claim 2,

$$\|\Delta W_\ell h_{\ell-1}(x)\|_2 = \|\Delta W_\ell\|_{\rightarrow} \cdot \|h_{\ell-1}(x)\|_2.$$

Using their initialization conditions (and similar scaling assumptions for  $\|\Delta W_\ell\|_{\rightarrow}$ ), they get that this first term already has size  $\Theta(\sqrt{n_\ell})$ . Unless it almost perfectly cancels with the other terms (which is unlikely in general), this gives a **lower bound**:

$$\|\Delta h_\ell(x)\|_2 = \Omega(\sqrt{n_\ell}).$$

Earlier, they used submultiplicativity and norm bounds to get a matching **upper bound**:

$$\|\Delta h_\ell(x)\|_2 = O(\sqrt{n_\ell}).$$

Putting upper and lower bounds together:

$$\|\Delta h_\ell(x)\|_2 = \Theta(\sqrt{n_\ell}),$$

matching the “desideratum” they were aiming for.

## 4. Concept-check questions

Here are some questions you can use to test your understanding of the arguments.

### Basic linear algebra & norms

The concept check questions for me seem fairly good, and something that could be a future HW problem about muP arguments. It is mainly recap to drill in the muP argument, which I think would be useful to help people remember the proof.

#### 1. Submultiplicativity.

Explain in your own words what the inequality

$$\|Av\|_2 \leq \|A\|_{\rightarrow} \|v\|_2$$
 means.

When can it be very loose? When is it tight?

#### 2. Semi-orthogonal case.

Suppose  $W \in \mathbb{R}^{m \times d}$  satisfies  $W^\top W = I_d$ .

- Show that all singular values of  $W$  are 1.
- Show directly that  $\|Wh\|_2 = \|h\|_2$  for any  $h$ .

#### 3. Rank-one matrix.

Let  $A = uv^\top$ . Prove that:

- $A$  has rank at most 1.
- $\|A\|_{\rightarrow} = \|u\|_2 \|v\|_2$ .
- For  $v$  itself,  $\|Av\|_2 = \|A\|_{\rightarrow} \|v\|_2$ .

### Random initialization

#### 4. Distribution of $Wh$ .

For  $W_{ij} \sim \mathcal{N}(0, \sigma^2)$  i.i.d. and fixed  $h$ , derive the distribution of each coordinate of  $Wh$ .

Then compute  $\mathbb{E} \|Wh\|_2^2$ .

#### 5. Scaling of $\|Wh\|$ .

Using the previous question and the law of large numbers, argue why  $\|Wh\|_2$  is concentrated around  $\sigma\sqrt{m} \|h\|_2$  when  $m$  is large.

#### 6. Comparing $\|Wh\|$ and $\|W\|_{\rightarrow} \|h\|$ .

Assume  $\|W\|_{\rightarrow} \approx \sigma(\sqrt{m} + \sqrt{d})$ .

Compute the ratio  $\frac{\|Wh\|_2}{\|W\|_{\rightarrow} \|h\|_2}$  and explain why it stays bounded away from 0 when  $m \geq d$ .

### Backprop and updates

#### 7. Gradient shape.

For a fully connected layer  $h_\ell = W_\ell h_{\ell-1}$ , derive the formula

$$\frac{\partial L}{\partial W_\ell} = \delta_\ell(x) h_{\ell-1}(x)^\top$$

starting from basic backprop rules.

### 8. Why rank-one?

Explain why, for batch size 1, the gradient update at a single layer is rank-one.

What happens if you use batch size  $B > 1$ ? What is the rank upper bound then?

### 9. Action on $h_{\ell-1}$ .

Starting from  $\Delta W_\ell = -\eta \delta_\ell(x) h_{\ell-1}^\top$ , compute  $\Delta W_\ell h_{\ell-1}(x)$  explicitly.

How does this relate to Claim 2?

## Scaling arguments

### 10. Feature norm induction.

Suppose you know  $\|h_{\ell-1}(x)\|_2 = \Theta(\sqrt{n_{\ell-1}})$  and  $\|W_\ell\|_{\rightarrow} = \Theta(\sqrt{n_\ell/n_{\ell-1}})$ .

Use Claim 1 to show that  $\|h_\ell(x)\|_2 = \Theta(\sqrt{n_\ell})$ .

### 11. Update norm bound.

Using the decomposition

$$\Delta h_\ell(x) = \Delta W_\ell h_{\ell-1}(x) + W_\ell \Delta h_{\ell-1}(x) + \Delta W_\ell \Delta h_{\ell-1}(x),$$

explain why it is reasonable (under generic conditions) to say that  $\|\Delta h_\ell(x)\|_2$  is at least on the order of  $\|\Delta W_\ell h_{\ell-1}(x)\|_2$ .

If you'd like, next step we can go through your own answers to some of these and I can give you feedback like I'm grading a CS182 solution.

Sources

Thanks! I'm still not completely clear on the proof of claim 2. Could you explain in more detail how  $\Delta h_\ell(x)$  and  $h_{\ell-1}(x)$  can be substituted for  $u$  and  $v$ , and how the observations you make apply to this case?

I wanted to test how well it would clarify questions I had, so I asked it this. The next explanation was very clear for me.

Great, let's zoom in on Claim 2 very concretely and literally match every symbol.

## 1. What Claim 2 is saying (in our notation)

We're at layer  $\ell$ , and for a **single training example**  $x$ :

- Input to the layer:  $h_{\ell-1}(x) \in \mathbb{R}^{n_{\ell-1}}$ .
- Weights:  $W_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$ .
- Output (pre-activation):  $h_\ell(x) = W_\ell h_{\ell-1}(x)$ .

Backprop gives us an **error signal** at this layer:

$$\delta_\ell(x) \in \mathbb{R}^{n_\ell}.$$

For batch size 1 and learning rate  $\eta > 0$ , the SGD update is

$$\Delta W_\ell = -\eta \frac{\partial L}{\partial W_\ell} = -\eta \delta_\ell(x) h_{\ell-1}(x)^\top.$$

Claim 2 says:

For this update matrix  $\Delta W_\ell$ , we have

$$\|\Delta W_\ell h_{\ell-1}(x)\|_2 = \|\Delta W_\ell\|_{\rightarrow} \|h_{\ell-1}(x)\|_2.$$

So the inequality  $\|\Delta W_\ell h_{\ell-1}\| \leq \|\Delta W_\ell\| \|h_{\ell-1}\|$  is actually an **equality** when you plug in the specific vector  $h_{\ell-1}(x)$ .

## 2. General linear algebra fact (rank-one outer product)

First recall the **general** fact we used (no neural nets yet):

Let  $A = uv^\top$  with

- $u \in \mathbb{R}^m$ ,
  - $v \in \mathbb{R}^d$ ,
- and neither vector is zero.

Then:

1.  $A$  has rank 1.
2. Its spectral norm is

$$\|A\|_{\rightarrow} = \|u\|_2 \|v\|_2.$$

3. If you multiply  $A$  by  $v$  (the same  $v$  used in the outer product), you get

$$Av = u \|v\|_2^2 \Rightarrow \|Av\|_2 = \|u\|_2 \|v\|_2^2 = \|A\|_{\rightarrow} \|v\|_2.$$

So **for this particular vector**  $v$ , the inequality

$$\|Av\|_2 \leq \|A\|_{\rightarrow} \|v\|_2$$

is tight: it holds with equality.

We'll now show that  $\Delta W_\ell$  is exactly of the form  $uv^\top$ , with  $u$  and  $v$  chosen from the neural-net symbols.

## 3. Matching the neural-net update to the rank-one form

You have:

$$\Delta W_\ell = -\eta \delta_\ell(x) h_{\ell-1}(x)^\top.$$

Compare this with the generic outer product:

$$A = uv^\top.$$

We can **literally identify**:

- $u := -\eta \delta_\ell(x) \in \mathbb{R}^{n_\ell},$
- $v := h_{\ell-1}(x) \in \mathbb{R}^{n_{\ell-1}}.$

Then:

$$\Delta W_\ell = uv^\top.$$

So everything we proved for a generic rank-one matrix  $uv^\top$  applies with this specific choice of  $u$  and  $v$ .

## 4. Re-running the rank-one argument with these symbols

### 4.1. Spectral norm of $\Delta W_\ell$

From the general fact:

$$\|\Delta W_\ell\|_{\rightarrow} = \|u\|_2 \|v\|_2 = \|-\eta \delta_\ell(x)\|_2 \|h_{\ell-1}(x)\|_2.$$

The minus sign doesn't matter for norms, so

$$\|\Delta W_\ell\|_{\rightarrow} = \eta \|\delta_\ell(x)\|_2 \|h_{\ell-1}(x)\|_2.$$

### 4.2. Action of $\Delta W_\ell$ on $h_{\ell-1}(x)$

Ignore this "action" phrasing, it is in this context a fancy way of saying "matrix multiplication". More generally, group "actions" are usually mentioned in group theory contexts (actions, orbits, etc.). I think these are sometimes used in quantum physics? Math 113, 114, 250 will talk more about them if you're interested

Now compute

$$\Delta W_\ell h_{\ell-1}(x).$$

Using the outer-product form:

$$\Delta W_\ell h_{\ell-1}(x) = uv^\top v = u(v^\top v) = u \|v\|_2^2.$$

Substituting back  $u = -\eta \delta_\ell(x), v = h_{\ell-1}(x)$ :

$$\Delta W_\ell h_{\ell-1}(x) = -\eta \delta_\ell(x) \|h_{\ell-1}(x)\|_2^2.$$

Now take the norm:

$$\|\Delta W_\ell h_{\ell-1}(x)\|_2 = \|-\eta \delta_\ell(x) \|h_{\ell-1}(x)\|_2^2\|_2 = \eta \|\delta_\ell(x)\|_2 \|h_{\ell-1}(x)\|_2^2.$$

### 4.3. Compare with $\|\Delta W_\ell\|_{\rightarrow} \|h_{\ell-1}(x)\|_2$

We already have

$$\|\Delta W_\ell\|_{\rightarrow} = \eta \|\delta_\ell(x)\|_2 \|h_{\ell-1}(x)\|_2.$$

Multiply by  $\|h_{\ell-1}(x)\|_2$ :

$$\|\Delta W_\ell\|_{\rightarrow} \|h_{\ell-1}(x)\|_2 = \eta \|\delta_\ell(x)\|_2 \|h_{\ell-1}(x)\|_2^2.$$

But that's **exactly** what we found for  $\|\Delta W_\ell h_{\ell-1}(x)\|_2$ . So

$$\|\Delta W_\ell h_{\ell-1}(x)\|_2 = \|\Delta W_\ell\|_{\rightarrow} \|h_{\ell-1}(x)\|_2.$$

This is Claim 2.

## 5. Intuition in words

- The update matrix  $\Delta W_\ell$  is built as an **outer product** of:
  - the backprop error at this layer,  $\delta_\ell(x)$  (up to a scalar  $-\eta$ ), and
  - the input vector to this layer,  $h_{\ell-1}(x)$ .
- Any outer product  $uv^\top$  has a unique “special” input direction  $v$  for which the matrix action is as strong as possible: if you feed in exactly this  $v$ , you get the maximal possible amplification, and the inequality  $\|Av\| \leq \|A\| \|v\|$  becomes equality.
- In our neural-net update, that special direction  $v$  **is the actual vector**  $h_{\ell-1}(x)$ . So the update  $\Delta W_\ell$  is perfectly aligned with this particular input: when you apply the updated layer to that same input, you see the full strength of the update’s spectral norm.

That’s all Claim 2 is saying: for batch size 1, the weight update is perfectly tailored to the particular training example’s incoming activation, so the “worst-case” direction for the update is exactly that activation.

If you’d like, next we can contrast this with **batch size**  $B > 1$ , where  $\Delta W_\ell$  becomes a sum of outer products and no longer rank-one, so you don’t get exact equality anymore—just an approximate alignment story.

Very small knitpick by me, just didn’t feel bringing in group-theoretic language was very helpful to the discussion.

This is a knitpick, but I saw you mentioned actions of  $\Delta W_\ell$  on  $h_{\ell-1}(x)$ . Is there a reason you brought in group-action language here? Do you think students in CS182 could benefit from looking at this operation as a group action, if so can you give a quick writeup that is accessible to them? If not, please choose different language

I didn’t *mean* “group action” in the formal algebraic sense there — I was just using casual language for “multiply the matrix by the vector.” So for CS182 purposes, it’s better to say:

“when we **apply**  $\Delta W_\ell$  to  $h_{\ell-1}(x)$ ”

or

“when we **multiply**  $h_{\ell-1}(x)$  by  $\Delta W_\ell$ ”

and avoid “action” unless we explicitly define group actions.